

Introduction

The aim of this study was to determine whether Labrasol® ALF, a non-ionic surfactant excipient, acts as intestinal permeation enhancer (PE), *ex vivo* and *in vivo*. Labrasol® comprises mono-, di- and triglycerides (Fig.1a) and mono- and di- fatty acid esters of PEG-8 (Fig.1b) and free PEG-8. PEs can be included in oral formulations to aid the transport of peptides such as insulin and GLP-1 across the intestinal epithelium through either the opening of tight junctions (TJs) or fluidisation of the plasma membrane. While Labrasol® ALF acts as a solubilizer, it can increase the oral bioavailability of active pharmaceutical ingredients. The mechanism by which it does this is not fully understood. Like many of the PEs identified to date such as sodium caprate, Labrasol® ALF has medium-chain fatty acid components within its structure.

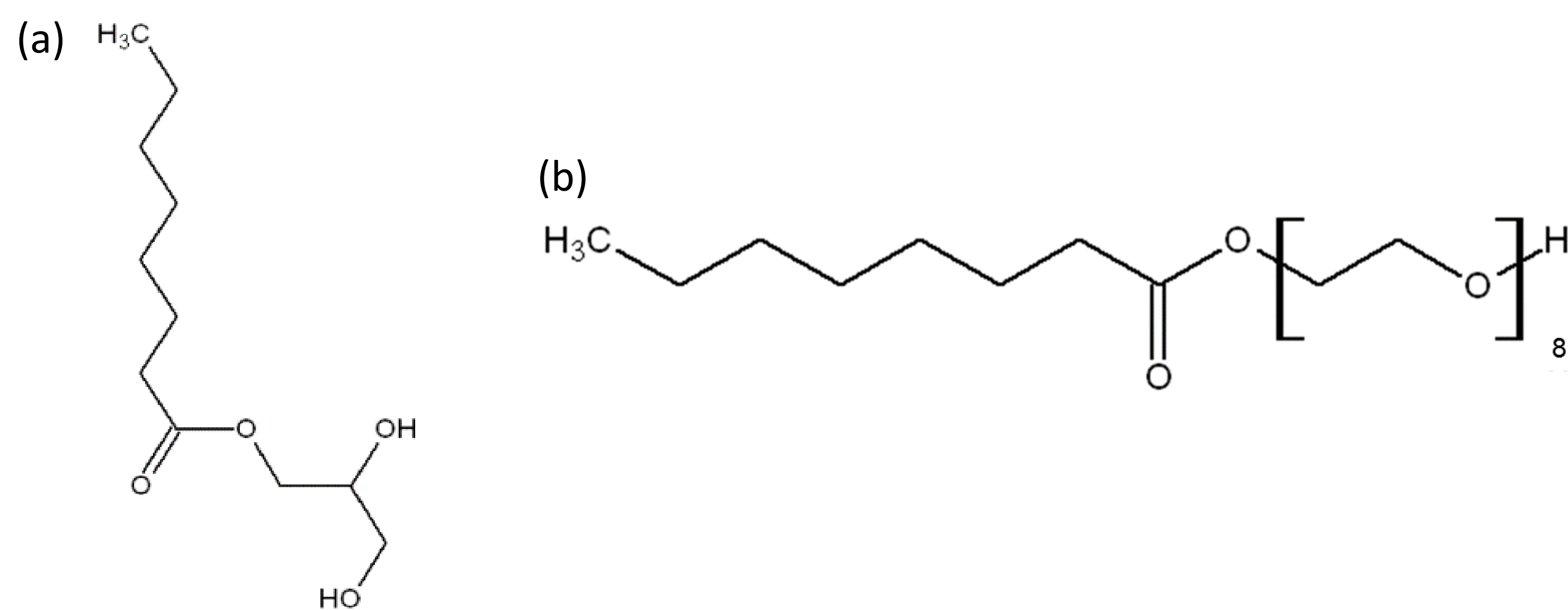


Figure 1: Components of Labrasol® ALF (a) Glycerol esters of caprylic (C₈) and C₁₀ acids and (b) PEG-8 esters of (C₈) and capric (C₁₀) acids¹

Methods

Labrasol® ALF (caprylocaproyl polyoxyl-8 glycerides) was tested in Ussing chambers using muscle-stripped rat colonic mucosae at concentrations of 2, 4, and 8 mg/ml on the apical side. Transepithelial electrical resistance (TEER, Ω) was calculated using Ohm's law. The apparent permeability (P_{app}) coefficients of [¹⁴C]-mannitol (0.1 μ Ci/ml) and a range of FITC-dextrans (4, 10, 20, 40 and 70 kDa, 2.5 mg/ml) were measured over 120 min. In situ jejunal instillations were performed in anaesthetised male non-diabetic Wistar rats. Either PBS, insulin (50 IU/kg) or Labrasol® (8 & 40 mg/ml) in combination with insulin were instilled using 30G needle. Blood was taken from the tail to measure blood glucose levels and retro-orbitally to measure plasma insulin using an ELISA (Mercodia).

Labrasol® reduces TEER and increases P_{app} of mannitol

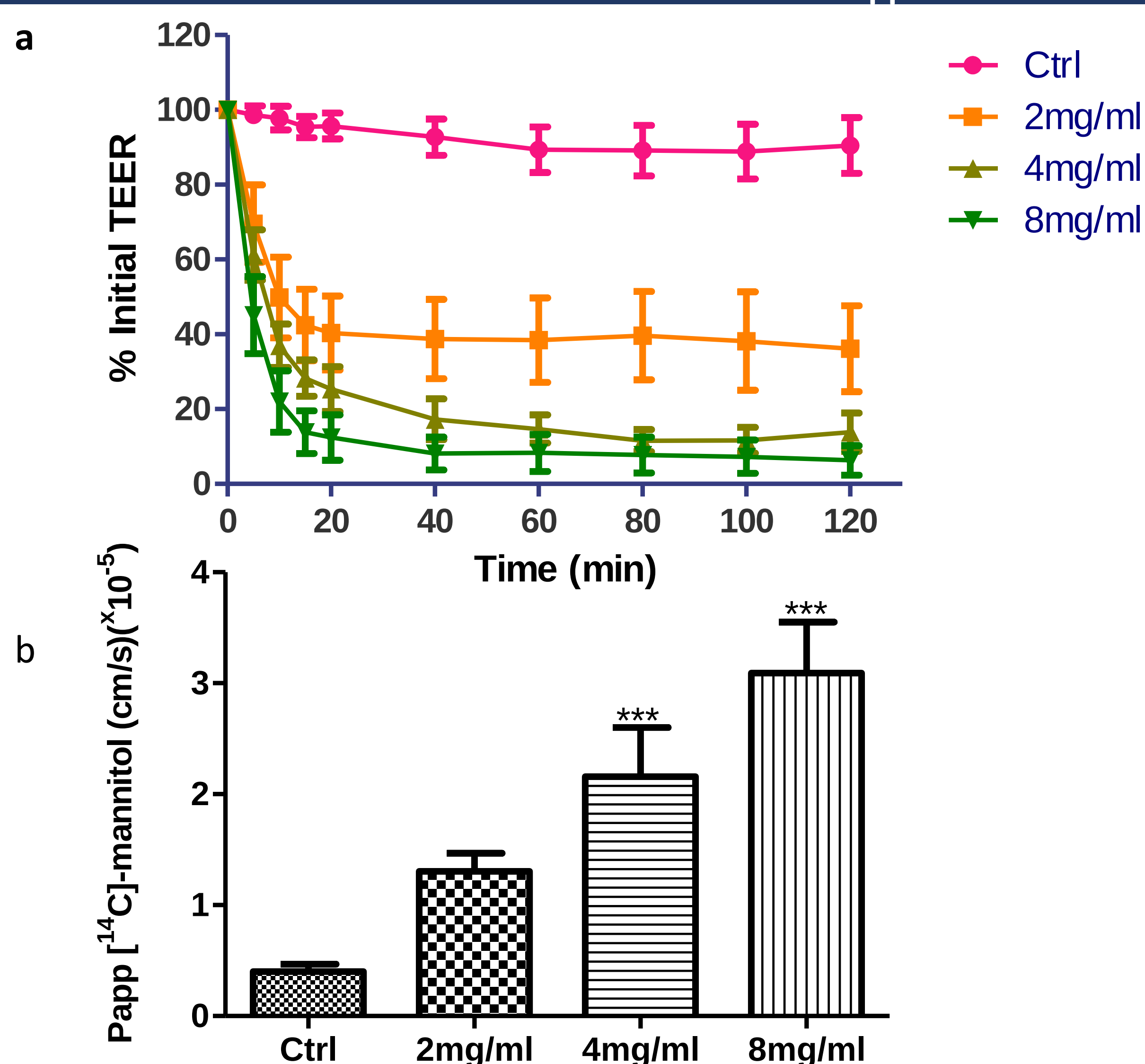


Fig. 2: Effect of Labrasol® (2-8 mg/ml) on (a) % initial TEER and (b) P_{app} of [¹⁴C]-mannitol in isolated colonic mucosae. Mean $\pm$ SEM, Control (Ctrl)(n=14) and test (n=8-9). *P<0.001 compared to untreated control, one-way ANOVA with Dunnett's *post-test*.**

A concentration-dependent decrease in TEER was observed after exposure to Labrasol® (2-8 mg/ml) to a maximum of 1% of initial TEER (Fig.2a). The basal P_{app} of [¹⁴C]-mannitol was 0.4×10^{-5} cm s⁻¹. Following exposure to Labrasol®, it increased to 1.3×10^{-5} cm s⁻¹ (2 mg/ml), 2.2×10^{-5} cm s⁻¹ (4 mg/ml), and 3.1×10^{-5} cm s⁻¹ (8 mg/ml) (Fig. 2b).

Labrasol® increases P_{app} of FITC dextrans in MW-dependent fashion

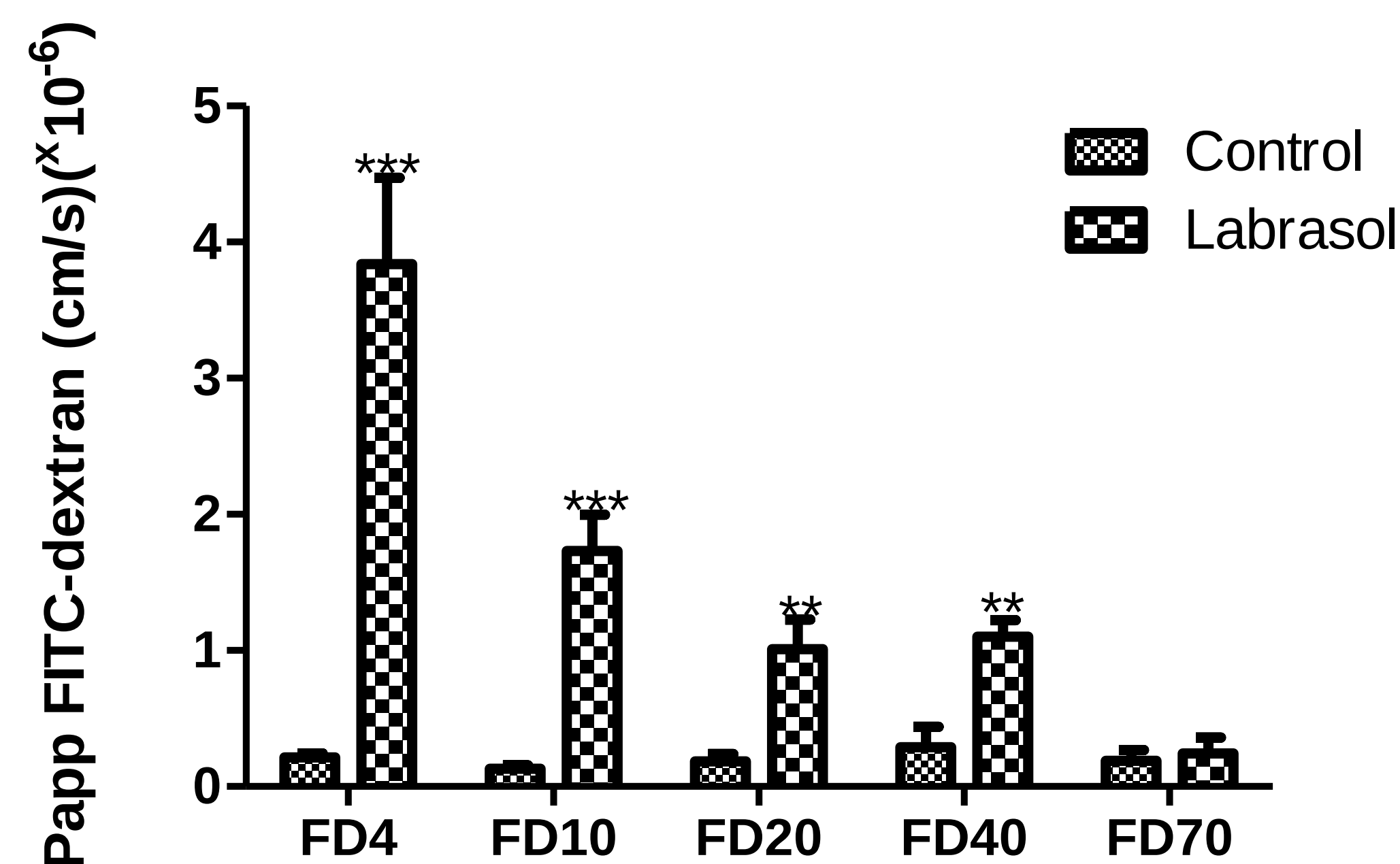


Figure 3: Labrasol® (8 mg/ml) increases the P_{app} of MW FITC-dextrans up to 40kDa across rat colonic mucosae. Mean $\pm$ SEM, n=4-5 (P<0.01, ***P<0.001 compared to untreated controls, unpaired Student t-test).**

In addition, Labrasol® (8 mg/ml) increased the P_{app} of FITC-dextrans in a molecular weight (MW)-dependent manner (Fig.3), but with no enhancement observed for FD70. This suggests that if the TJs are opening, they only allow polar molecules of certain dimensions and MW to pass through and not 'bystanders' which typically have MW > 70kDa [2].

Labrasol® decreased blood glucose levels and increased plasma insulin levels

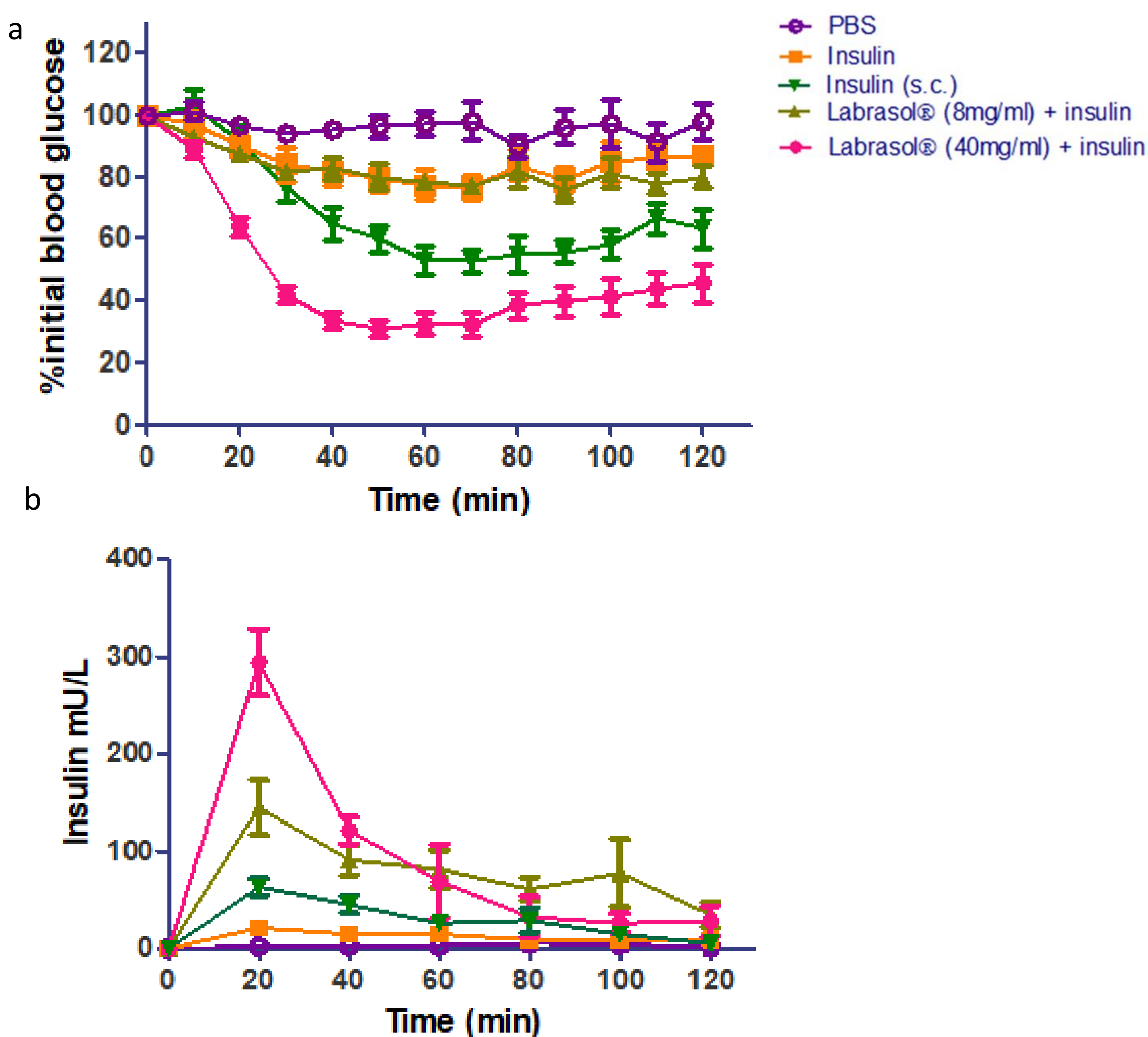


Figure 4: (a) Reduction in blood glucose (b) increase in plasma insulin (mU/L) after in situ instillation of Labrasol® (8 & 40 mg/ml) in combination with insulin (50 IU/kg) in rat jejunum. Insulin and PBS were administered as controls. Insulin (1 IU/kg) was administered sub cutaneously (s.c.) in order to calculate relative bioavailability. Mean $\pm$ SEM and n=6.

Administration of Labrasol® ALF (8, 40 mg/ml) with insulin (50 IU/kg) to rat jejunum using the in situ instillation technique, over 120 min, induced a decrease in blood glucose levels and an increase in plasma insulin levels compared to PBS or insulin alone. Insulin ad-mixed with Labrasol® ALF induced a relative bioavailability of 5% (8 mg/ml) and 7% (40 mg/ml) compared to subcutaneous administration of 1 IU/kg of insulin.

Conclusion

This data suggests that Labrasol® ALF acts as an intestinal PE likely by the opening of TJs and could be included in formulations for peptides to improve oral bioavailability.

References

- Jannin, V. et al. International Journal of Pharmaceutics, 2014. 466: 109-121
- Tuvia, S., et al., Pharm Res, 2014. 31: 2010-21.